EE6110 Special Topics in Automation and Control (Advanced)
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SafeGrasp: Occlusion-Aware Safe Grasping in Clutter with Control Barrier Functions

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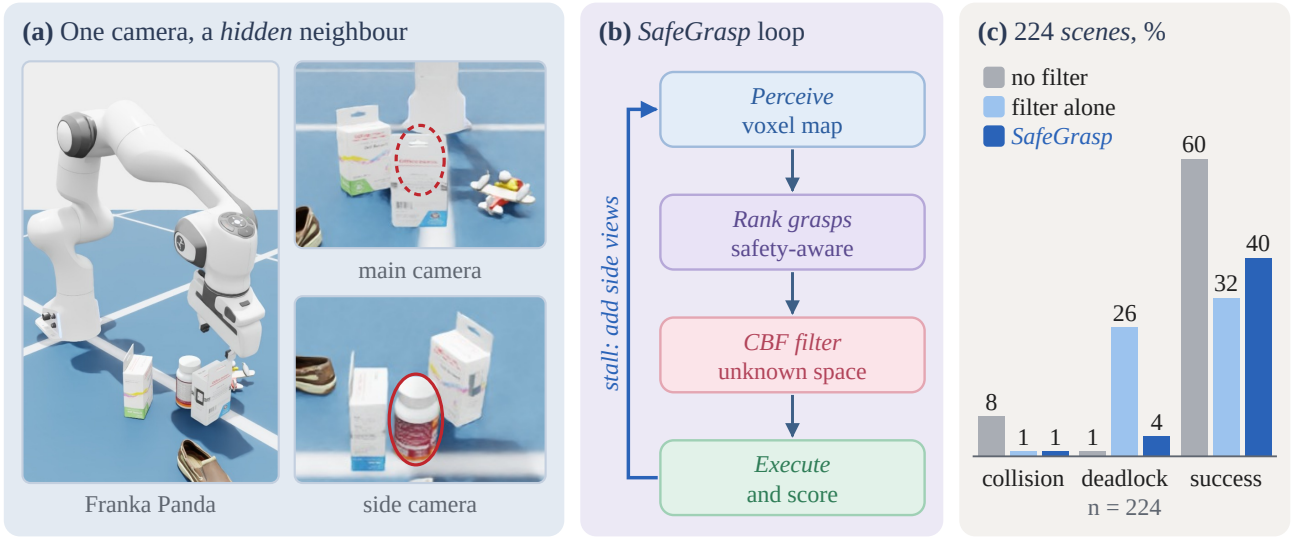


Fig. 1: Overview of SafeGrasp. (a) Isaac Sim renders of HIDDEN scene (the scene of Fig. 6): the Franka Panda reaching for the target box, and the main and side camera views. The bottle behind the target (red ring) is invisible to the main camera. (b) The stages of SafeGrasp. On a stall, the scene is re-observed with two side cameras. (c) On the 224 benchmark scenes, SafeGrasp collides in 1% and deadlocks in 4% (separate batch, same scenes).

1 Introduction

A robot that picks a single object from a cluttered table must avoid the objects around it. A single depth camera observes only front surfaces, so the space behind every object remains unknown. If the planner treats this unknown space as free, the arm can strike an object it has never observed. However, if it treats it as solid, the arm often cannot reach the target.

Grasp detectors are evaluated on grasp success, pile clearance and speed [2, 3, 5, 22], and not on what the arm hits on the way. Control barrier functions (CBFs), i.e., constraints that limit how fast the arm may approach an obstacle so that it never enters it, keep manipulators out of obstacles [7, 8], but only out of those the camera has observed.

In this work, we propose SafeGrasp, a safety layer for grasping in clutter from a single depth camera. It adds a soft barrier on unknown space to a CBF filter, exempts the target's own shadow, ranks grasps by safety, and re-observes the scene with two side cameras when the arm stalls. We compare it against the unfiltered antipodal grasp (`ant_nom`), a potential field (`ant_apf`), a plain CBF (`ant_cbf`) and our filter without recovery (`safe_oacbf`). The main findings are as follows. First, the filter alone trades success for safety: collisions fall from 8% to 1%, but success falls from 60% to 32% because 26% of trials stall before the grasp (1792 paired trials). Second, SafeGrasp recovers part of this loss: re-observing on a stall reduces the deadlock to 4% and raises success to 40%, with 1% collisions. Third, observation coverage matters more than controller tuning: two side cameras raise the filter from 24% to 55% success, whereas no gain or margin setting produces a significant change. Fourth, with Segment Anything masks the robot picked 6 of 10 objects and carried all 6 to a basket detected from RGB-D. Finally, success saturates near 60% because the close fails, and none of the fixes we tried overcame this.

Section 3 describes the baseline system, and Sections 4 to 6 the randomised trials and the four extensions: heavy occlusion, contact-aware safety, failure recovery and learned perception.

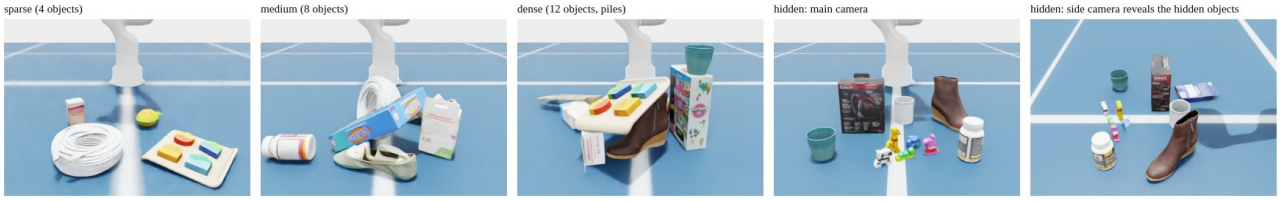


Fig. 2: The four clutter levels from the main camera, and one HIDDEN scene from the main and side cameras, where the small objects behind the tall box are invisible to the main camera.

2 Related Work

Analytic grasp synthesis reasons about force closure and antipodal contacts [1], and GPD samples and scores grasps in point clouds [2]. Learned methods predict grasp quality from a depth image, as in GG-CNN [3, 4], or from a volumetric map, as in VGN [5]. On real robots, these detectors report grasp or pile-clearance rates of 84 to 93% [2, 3, 5, 22]. Those figures are not directly comparable with ours. The detectors grasp whichever object in the pile scores highest, and pile clearance is counted over repeated grasps, whereas our trials demand a specified target, with no freedom to choose the easiest object, and are scored in simulation, where the close failures analysed in §5 cap success near 60%. Potential fields push the arm away from obstacles but can trap it in local minima [6]. CBFs render a safe set forward invariant through a small quadratic program [7] and have been used as kinematic safety filters for manipulators [8]. Our filter follows this kinematic formulation and adds a soft barrier on unknown space derived from a voxel map, in a similar way to occupancy mapping [9].

3 System Design

In this work, the main proposed method, SafeGrasp, is the configuration `safe_oacbf`, i.e., the safety-aware ranking (`safe`, §3.3) with the occlusion-aware filter (`oacbf`, §3.4) and the target-shadow carve (§3.2), plus stall-triggered re-observation (§3.4). All other configurations are baselines.

3.1 Workcell, scenes and pipeline

A Franka Panda [10] with a parallel-jaw gripper faces a table in Isaac Sim [11, 12]. A single main depth camera observes the robot from across the table at about 21° elevation, and side and wrist cameras are used only where stated. The objects come from the Google Scanned Objects dataset [13] with convex collision meshes [14]. Each scene has one of four clutter levels (Fig. 2) and is settled by physics before the camera captures it: SPARSE scenes contain 4 objects, MEDIUM 8, and DENSE 12 in piles. In HIDDEN scenes, the target is a tall object at the front with one or two short neighbours flush behind it in its camera shadow, where the far finger must pass.

Every trial follows the stages in Fig. 3. Only the proposer and the executor differ between methods, so each method is named `{proposer}_{executor}`, e.g., `ant_nom`, and the benchmark evaluates eight such combinations (Table 2).

3.2 Perception and the unknown-space model

Depth images are back-projected into a world-frame point cloud and voxelised (Fig. 4). A voxel is either *occupied* (target or obstacle, labelled from instance masks) or *unknown*, i.e., no camera ray has reached it, and a Euclidean distance transform gives the distances d_{obs} and d_{unk} to the two classes. The target's own shadow is its hidden rear surface, exactly where the far finger must be placed. Hence, unknown voxels within `carve_radius` (2 cm) of the target's visible surface are *carved* free, otherwise the barrier would push the hand off the object being grasped. In §6.4, Segment Anything masks replace the ground-truth masks to demonstrate a real-world implementation.

3.3 Grasp proposers

A proposer takes the partial target cloud and returns ranked 6-DoF grasps with a commanded width.

- `td`: vertical grasps aligned with the target's principal axis.
- `ant`: antipodal grasps sampled on the cloud from surface normals, our strongest analytic proposer.
- `safe`: antipodal grasps re-ranked by clearance. Inverse kinematics (Levenberg–Marquardt [15]) yields the pre-grasp and grasp, the arm's collision spheres are checked along the transit and approach, and a grasp is demoted if its approach enters the obstacle margin or its corridor enters the unknown-space margin.
- `ggcnn`: GG-CNN2 [4] with Cornell-trained weights [16], applied to a top-down heightmap of the observed cloud.

The analytic proposers also require at least 20 observed target points between the pads, spread over most of the pad, so that both pads make contact.

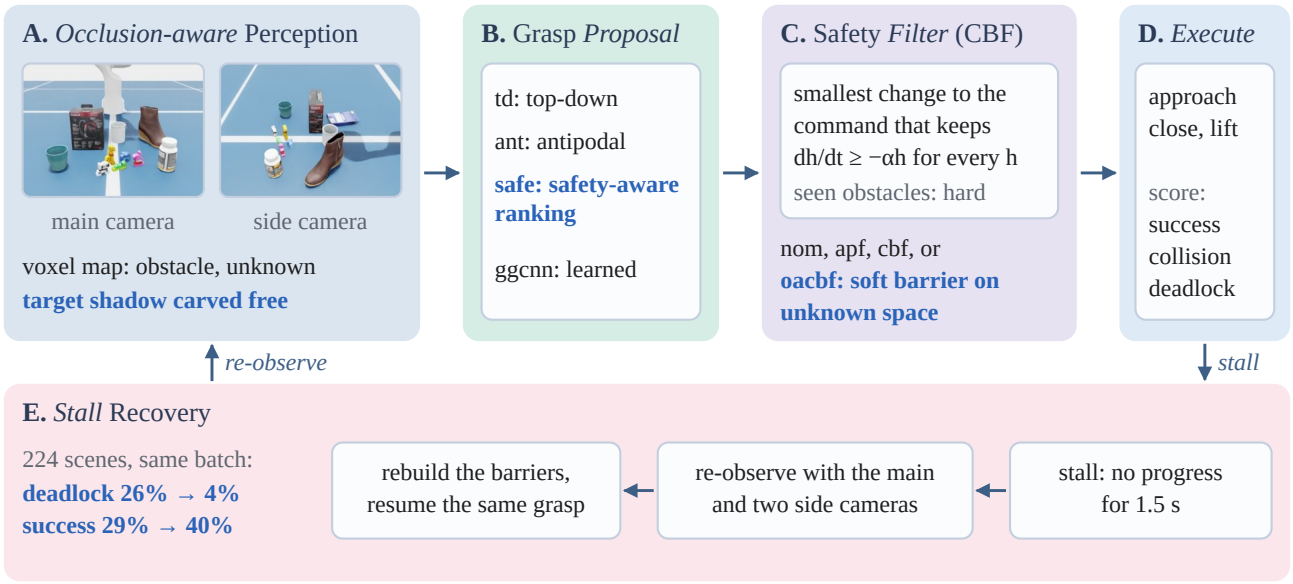


Fig. 3: Stages of one trial. A: one or three depth cameras build a voxel map of obstacle and unknown space, with the target's shadow carved free. B: a proposer ranks grasps. C: the CBF filter modifies the command as little as possible. D: the arm approaches, closes, lifts and is scored. E: on a stall, SafeGrasp re-observes with the side cameras and resumes the same grasp. SafeGrasp's choices are in bold blue.

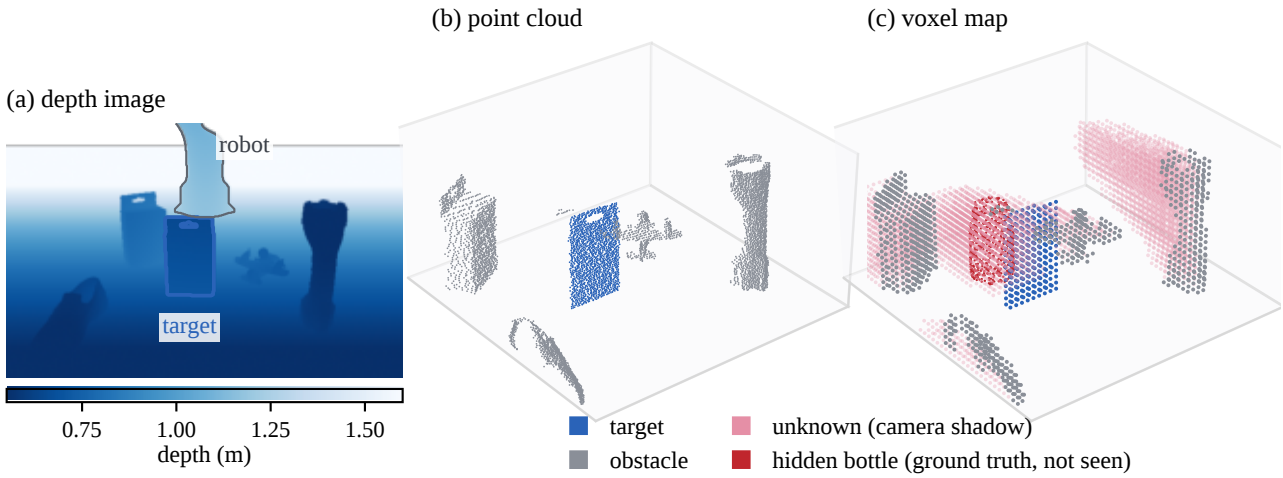


Fig. 4: Perception on HIDDEN scene from the main camera alone. (a) The depth image, with the target (blue) and the robot (grey) outlined from the instance masks. (b) The point cloud: target points in blue, obstacle points in grey. (c) The voxel map: target and obstacle voxels, and the unknown voxels in the camera's shadow (pink) that form the soft barrier. The hidden bottle (red, ground truth) lies inside that shadow, whereas the target's own shadow is carved free.

3.4 Safety-filtered execution

The nominal controller drives the tool centre point to the pre-grasp and then the grasp by resolved-rate control with a damped-least-squares inverse [17]. A safety executor may filter the nominal joint velocity $\dot{q}_{\text{nom}}$, passing it through when it is safe and modifying it as little as possible near a barrier (Fig. 5). The arm and fingers are covered by spheres with centres $p_i(q)$, radii r_i and Jacobians $J_i(q)$. For sphere i and barrier b (obstacle or unknown), the barrier function is $h_{ib} = d_b(p_i(q)) - r_i - \delta_b$ with safety margin δ_b, and the filter keeps $h \geq 0$ by enforcing $\dot{h} \geq -\alpha h$ with gain α in $1/\text{s}$. Since $\dot{h} = \nabla d_b^{\top} J_i \dot{q}$ is linear in $\dot{q}$, the filter reduces to the quadratic program, as follows:

$$\begin{aligned} \dot{q}^{\star} = \arg\min_{\dot{q},\, s} \; & \|\dot{q} - \dot{q}_{\text{nom}}\|^2 + \rho\|\dot{q}\|^2 + w_s \textstyle\sum_{\text{soft}} s_{ib}^2 \\ \text{s.t. } & \nabla d_b^{\top} J_i \, \dot{q} + s_{ib} \geq -\alpha \, h_{ib}, \end{aligned} \tag{1}$$

with joint and velocity limits as box constraints and a small damping weight ρ. Soft barriers receive a slack $s_{ib} \geq 0$ with penalty weight w_s, so that they can be violated at a cost without making the problem infeasible. If it is still infeasible,

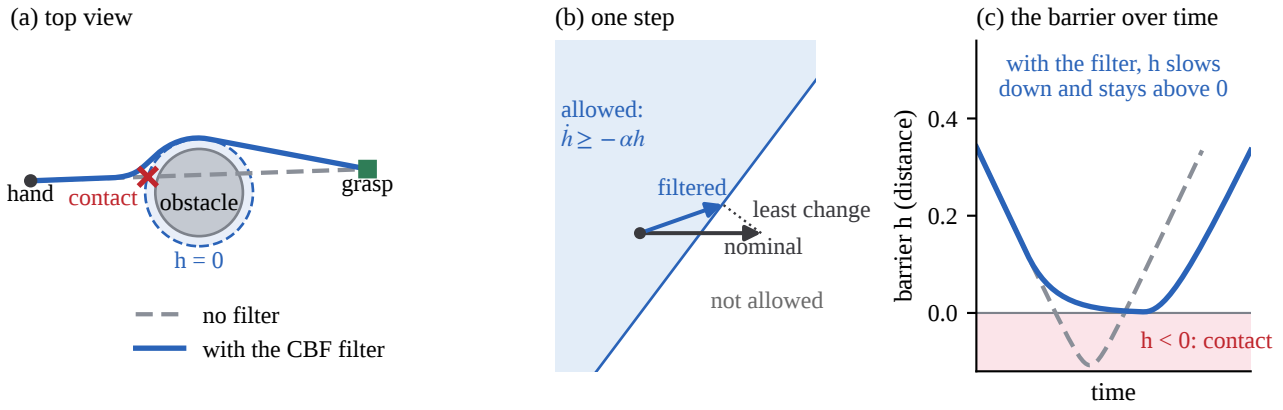


Fig. 5: Illustration of a CBF filter on a toy example. (a) A hand moves towards a grasp behind an obstacle. The barrier h is the distance to the obstacle minus the hand radius, so $h = 0$ on the dashed ring. Without a filter, the hand collides. (b) At each step the filter admits only commands with $\dot{h} \geq -\alpha h$ (blue side) and selects the admissible one closest to the nominal command. (c) Hence, h may decrease, but ever more slowly near zero, and never crosses it. SafeGrasp applies this rule to every arm sphere, with a second, soft barrier on unknown space.

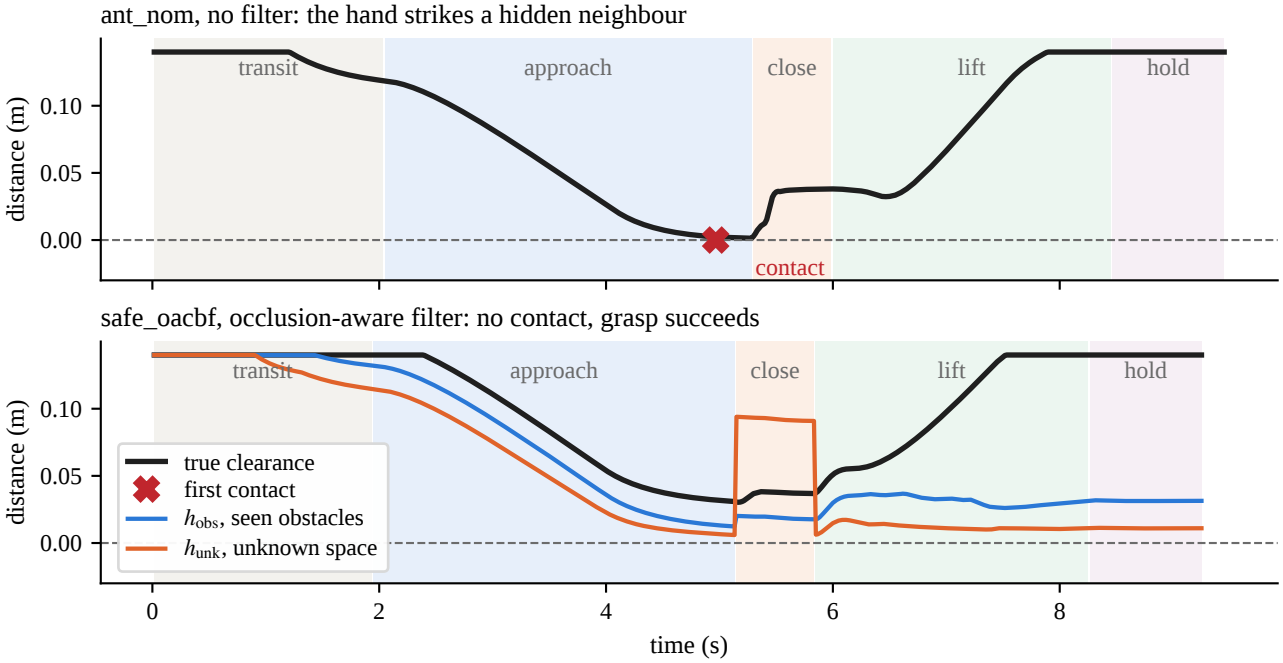


Fig. 6: The occlusion-aware filter on a HIDDEN scene. Top: without a filter (`ant_nom`), the true clearance falls to about a millimetre and the hand touches a hidden neighbour (red cross). Bottom: with `safe_oacbf`, h_{obs} and h_{unk} remain positive and the grasp succeeds without contact. The unknown-space barrier is relaxed while the fingers close.

constraints are relaxed in a fixed order. DAQP [18] solves it in 0.4 ms or less per control step on average over the benchmark. The four executors are `nom` (no filter), `apf` (a repulsive velocity away from observed obstacles [6]), `cbf` (Eq. 1 with the obstacle barrier only) and `oacbf`, which adds the soft barrier on unknown space outside the carved target shadow (Fig. 6).

The final component of SafeGrasp is stall recovery: when the arm moves less than 2 mm in 1.5 s under `oacbf`, the robot re-runs perception with the main and two side cameras, rebuilds the barriers and resumes the same grasp, at most twice per trial. The side cameras are not used from the start because the problem setting is grasping from a single camera. Re-observing on demand keeps that setting in the trials that never stall, which are the majority, and pays for the extra views only where the main camera is insufficient. With the side cameras always on, the occlusion problem largely disappears, and so does the role of the safety filter (§6.1).

Table 1: Proposed method against baselines on the 224 benchmark scenes, rates in %. Top: the benchmark batch. Bottom: SafeGrasp and the filter alone, re-run together in a separate batch on the same scenes. Only totals are comparable across batches, since the scenes settle slightly differently (the target matched in 177 of 224).

method	collision	deadlock	success
`ant_nom` (no filter)	8	1	60
`ant_cbf` (plain CBF)	5	1	56
`safe_oacbf` (filter alone)	1	26	32
`safe_oacbf`, re-run	1	26	29
SafeGrasp (filter + re-observe)	1	**4**	**40**

4 Experimental Setup

Platforms. The main benchmark, the controller ablation and the recovery run (Sections 5 to 6.2) were run on Isaac Sim 5.1 with Isaac Lab 2.3.2. The later experiments on the close and on learned segmentation (§6.3 and §6.4) were run on Isaac Sim 6.1 with Isaac Lab 3.0, in MEDIUM clutter and with ground-truth masks unless stated otherwise, because the older renderer crashed on the machine used for them. The newer Isaac Lab no longer exposes two PhysX contact settings that the benchmark runs tuned, so rates are not comparable across the two platforms, and every comparison in this report is paired within one platform against a baseline run on the same scenes.

Trials. The benchmark has 224 scenes (60 sparse, 57 medium, 48 dense and 59 hidden) with 436 distinct objects and 8 methods, i.e., 1792 trials. The controller ablation re-runs 51 of these scenes (22 dense, 29 hidden, 252 distinct objects) for 18 variants, i.e., 918 trials, and the recovery run repeats the same 51 scenes (253 distinct objects). The later experiments use 15 objects.

Metrics. *Success* means no execution failure and a target ending more than 8 cm above its start. The later experiments also require the gripper to end more than 2 mm open, which would change at most 4 of 224 trials per method in the benchmark. *Collision* means a robot link or finger touches a non-target object with more than 1 N. *Disturbed* means a neighbour moved more than 2 cm or rotated more than 15°, and *clean success* is a success with neither. *Deadlock* means every attempt stalled or timed out before the close, a *slip* is a lift that leaves the target below 8 cm, and an *empty close* ends with the fingers closed on nothing.

Statistics. Each scene is replayed for every method, so methods are compared on the same scenes. A difference is called significant when $p < 0.05$ after correction for the number of comparisons. Smaller differences lie within run-to-run noise: in our null control, 7 of 24 scenes changed outcome between two identical runs.

5 Main Results

SafeGrasp is the filter alone (`safe_oacbf`) plus stall-triggered re-observation with the side cameras (§3), which is the last row of each table below.

SafeGrasp is the only configuration with at most 2% collisions and less than 5% deadlock (Tables 1 and 2, $n = 224$). It gives up 20 percentage points of success relative to the unfiltered grasp, which collides in 8%, and gains 11 points over the filter alone in the same batch, a significant difference. Its deadlock (4%) and that of the unfiltered grasp (1%) differ only within run-to-run noise.

Without recovery, the filter achieves its low collision rate at the cost of deadlock (Table 2). It has the fewest collisions but succeeds in only 32% against 60%, because it treats unknown space as an obstacle and finds no admissible path in 26% of trials. The safety-aware ranking lowers the deadlock rate from 44% (`ant_oacbf`) to 26%, a difference just within noise. The plain CBF retains 56% success with 1% deadlock, whereas the potential field drops to 34% with 38% deadlock because its repulsion traps the arm.

The trade-off is most pronounced on the hidden scenes ($n = 59$), where `ant_nom` reaches 81% but collides in 5 of 59 because its far finger enters the shadow in which the neighbours sit, and `ant_oacbf` has no collisions but deadlocks in 80% because its 2 cm carve is narrower than the reach behind the target. Across all scenes, the filters trade collisions for deadlock, whereas the other methods fail mostly by slips (Fig. 7).

Most of the remaining failures are empty closes: 83 of the 90 `ant_nom` failures are slips, and in the later experiments (§6.3) the gripper closes on nothing in these cases. The object appears to be pushed out during the close rather than missed, since re-centred grasps with far more observed points between the pads fail equally often (§6.3). The failures concentrate on

Table 2: Main benchmark, $n = 224$ paired scenes per method (1792 trials), rates in %. Right: success by clutter level. Bold: best in each column. SafeGrasp is the filter plus re-observation on a stall, run in a separate batch on the same scenes, in which the filter alone reached 29% success and 26% deadlock.

	all 224 scenes					success by clutter level			
method	success	collision	disturbed	clean success	deadlock	SPARSE $n = 60$	MEDIUM $n = 57$	DENSE $n = 48$	HIDDEN $n = 59$
`safe_oacbf` (filter alone)	32	**1**	17	22	26	37	32	29	29
`td_nom`	47	16	21	33	22	53	42	**42**	51
`ant_nom`	**60**	8	22	**46**	1	**62**	**51**	**42**	**81**
`ant_apf`	34	3	**10**	29	38	45	21	8	58
`ant_cbf`	56	5	19	43	1	**62**	42	40	78
`ant_oacbf`	25	2	13	19	44	35	32	25	10
`safe_nom`	50	4	21	38	**0**	53	47	38	61
`safe_cbf`	48	3	21	36	**0**	52	42	33	63
SafeGrasp (ours)	40	**1**	16	34	4	38	35	31	53

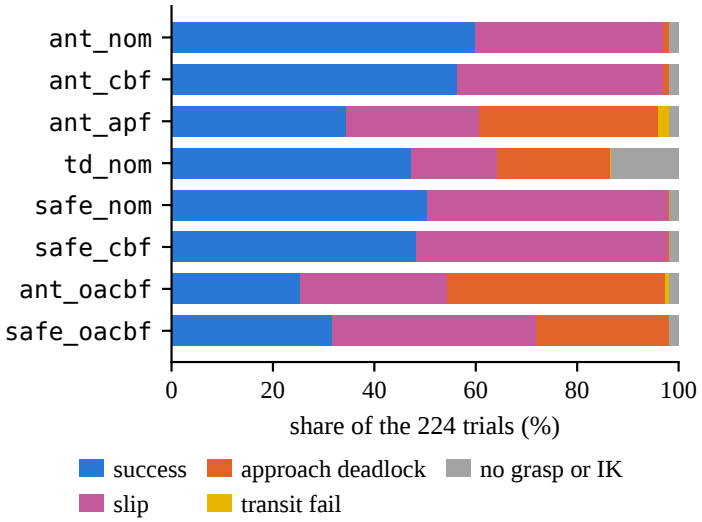


Fig. 7: Outcome of every benchmark trial by method ($n = 224$ each). The occlusion-aware executors convert collisions into approach deadlock (orange), while most other failures are slips (pink).

difficult objects such as thin Jenga blocks, a phone and a shoe. Pooled over the later experiments, the baseline succeeds in 55% (97 of 177 trials), and two identical runs change outcome on 7 of 24 scenes, which bounds the effect any intervention can demonstrate.

6 Extensions and Ablations

6.1 Controller ablation

Observation coverage has by far the largest effect (Table 3, $n = 51$). Two side cameras raise `safe_oacbf` from 24% to 55% success ($p = 0.002$) and reduce deadlock from 33% to 2%, because the additional views convert unknown voxels into known free space. On the 29 hidden scenes, success rises from 24% to 79%. With permanently active side cameras the unfiltered grasp also had no collisions, i.e., full coverage removes the occlusion problem rather than handling it, which is why SafeGrasp activates the side cameras only when the arm stalls (§6.2).

None of the controller parameters produces a significant change. A wider carve (4 cm) helps most, raising success to 39% and reducing deadlock to 12%, whereas a tighter unknown-space margin raises deadlock to 59%. Tripling the depth noise reduces deadlock to 8% only because the dropped edge pixels shrink the shadow, while the unfiltered grasp falls from 65% to 49%.

6.2 Deadlock detection and recovery

Re-observing the scene when the arm stalls removes almost all deadlocks (Table 4, $n = 51$). Deadlock fell from 12 to 1 of 51 ($p = 0.002$) and success rose from 12 to 22, both significant, with no collisions, because the side views shrink the

Table 3: Controller ablation, $n = 51$ paired dense and hidden scenes per row. One parameter changes per row. Rates in %. †: significant difference in success from the `safe_oacbf` reference, with correction over all 17 comparisons.

variant	success	collision	deadlock
`safe_oacbf` (reference)	24	0	33
`alpha=2`	29	2	33
`margin_unk=0.0`	33	0	22
`margin_unk=0.015`	16	0	59
`carve_radius=0.0`	22	0	39
`carve_radius=0.04`	39	0	12
`noise=3` (×3 noise)	29	0	8
3 views (main, left, right)	**55**†	0	2
`ant_nom`, 3 views	67†	0	0
`ant_nom`, `noise=0`	65†	8	2
`ant_nom`, `noise=3`	49	0	0

Table 4: Deadlock recovery, $n = 51$ paired dense and hidden scenes per arm, i.e., the scenes of Table 3 in a new run. Rates in %. †: significant difference from A0 and A2.

arm	success	collision	deadlock
A0 filter alone (`safe_oacbf`)	24	0	24
A1 re-observe (SafeGrasp)	**43**†	0	**2**†
A2 relax the margin on stall	24	0	24

unknown-space barrier to a median of 13% of its voxels (13 recoveries). The largest gain was on the hidden scenes (10 to 18 of 29), and the trials did not take longer (mean 11.8 against 11.5 s). The full 224-scene run confirms this result (Table 1).

Relaxing the unknown-space margin on a stall instead changes no outcome, because the arm then stops at the boundary of the unknown region itself. Hence, the deadlock stems from what the camera cannot see rather than from the margin. On-demand views match the deadlock rate of permanently active side cameras but not their success (22 against 28 of 51), likely because the grasp is still selected from the main view. Only totals are comparable across runs (the reference deadlocked in 12 of 51 here and 17 of 51 in Table 3), and the side images were captured with the arm at its home pose, so a real robot would also see its own arm.

6.3 Execution and the close

Table 5: Interventions on the close, each paired against the `ant_nom` baseline on the same scenes (raw success). The null control runs the baseline against itself. The proposer and re-plan rows are 15-scene pilots without raw logs (team record only). No row shows a significant difference.

intervention	n	base → new	note
GG-CNN2 proposer	15	8 → 3	9 → 4 with the pad filter
re-plan, always	15	10 → 5	swapped good grasps for worse
re-plan, gated	15	10 → 8	fixed 3 slips, broke 5 successes
empty-close recovery	24	14 → 13	re-centred grasps still slip
soft close (1.8 s)	60	30 → 30	no effect
null control (twice)	24	13 → 16	7 of 24 scenes flip

GG-CNN2 did not outperform the analytic antipodal proposer (Table 5, $n = 15$), succeeding in 3 of 15 scenes against 8, although the pilot is too small to estimate the gap. It predicts top-down grasps only, which already trailed antipodal grasps in the benchmark (47% against 60%), and it receives a synthetic heightmap although trained on real depth, so it stalled on the approach in 6 of 15 scenes. We also integrated VGN [5] but could not verify how its grasps convert to our gripper frame, so its 15 failed trials are excluded.

None of the interventions on the close produced a significant difference, and the null control explains why: running the baseline twice changed the outcome of 7 of 24 scenes, more than the effect of the empty-close recovery (one scene) or the 1.8 s soft close (none). A wrist-camera re-plan halved success when it always adopted the new grasp, and a gated version still scored 8 against 10, because the wrist capture itself steps the physics and the trial drifts from its open-loop counterpart.

Empty-close detection works, but the recovery does not raise success ($n = 24$). A gripper width of 6 mm or less flags an empty grasp, after which the robot re-centres on the target and retries. The recovery triggered in 7 of 24 trials and rescued

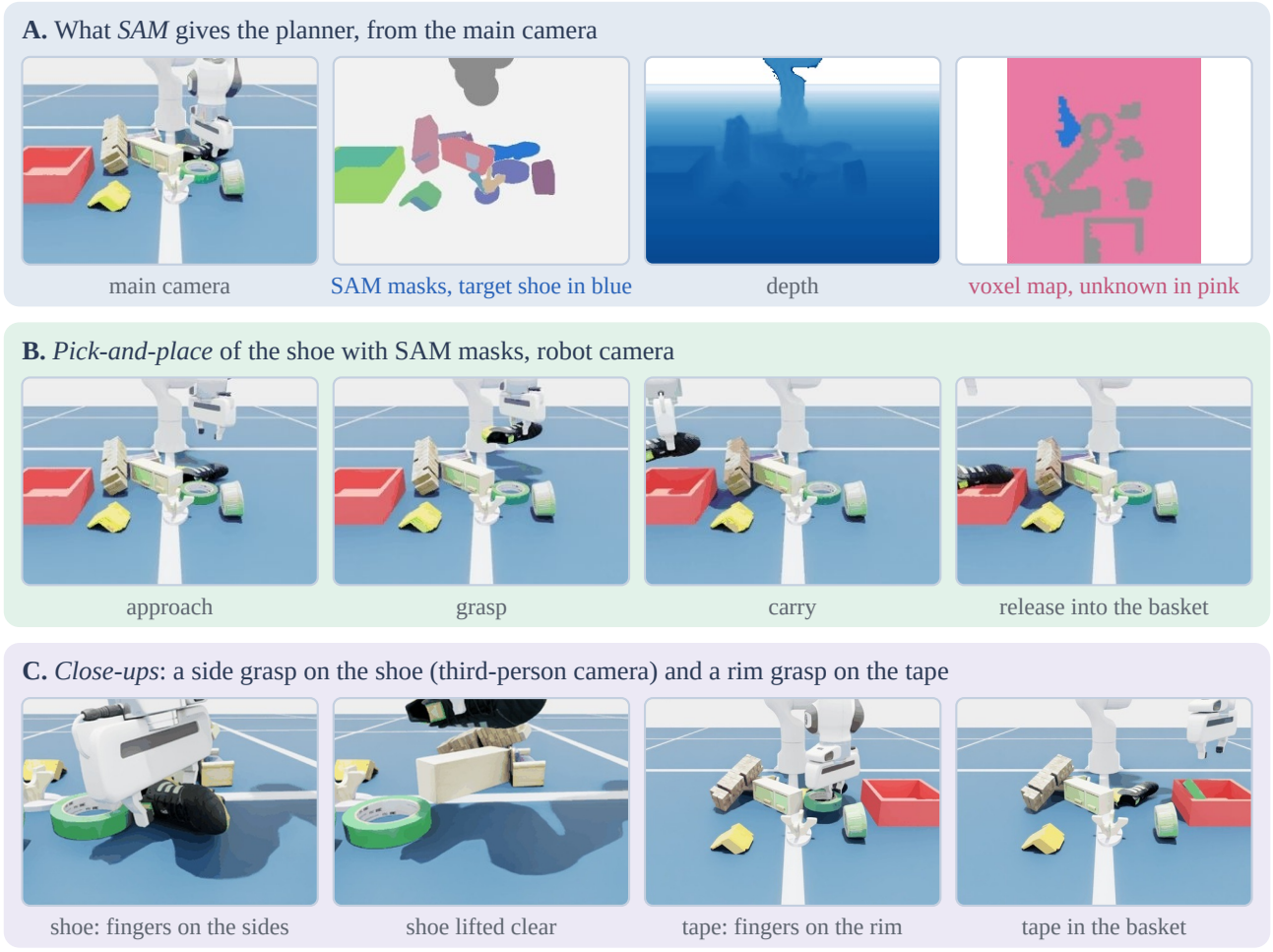


Fig. 8: SAM on one medium demonstration scene. A: the main camera image, the SAM masks (target shoe in blue, every other mask an obstacle), the depth image, and the top-down voxel map built from them (grey: obstacle, pink: unknown). B: the shoe picked and released into the basket detected from RGB-D. C: a side grasp on the shoe from a third-person camera (ground-truth masks), and a rim grasp on a roll of tape with SAM masks, carried to the basket.

1. The retried grasps placed 365 to 828 observed points between the pads, against 121 to 549 before, yet the object still escaped.

6.4 Learned segmentation and pick-and-place

With learned masks, the robot picked 6 of 10 objects and carried all 6 to the detected basket ($n = 10$ unpaired seeds, `ant_nom`). Segment Anything [21] (ViT-B, main camera, CPU) replaces the ground-truth masks, every SAM region except the target is treated as an obstacle, and the drop point is a red basket detected each run from colour and depth (Fig. 8). The target is still selected from the renderer's labels, standing in for a user pointing at it, and the same loop handles other shapes, e.g., a rim grasp on a roll of tape (Fig. 8C).

Basket detection was the only source of placement error: it located the basket within 1 cm in 7 of 10 scenes but was 19 and 23 cm off in 2 where it was partly hidden, so about 5 of 10 objects ended inside the real basket (inferred, since only arrival at the detected centre was logged). These runs show that SAM masks can drive the full loop, but they are not a benchmark: the seeds are unpaired, a pick requires only a 3 cm rise and the empty-close retry was enabled.

7 Conclusion

We have proposed SafeGrasp, which reduces collisions to about 1% at a cost of 20 percentage points of success: on the 224 benchmark scenes it collides in 1%, deadlocks in 4% and succeeds in 40%, against 8%, 1% and 60%, respectively, for the unfiltered grasp. Stall-triggered re-observation is the decisive component, since it reduces the filter's deadlock from 26% to 4%, and the ablation attributes the filter's loss of success to missing observations: additional views recover it, whereas no gain or margin setting does. We hypothesise that a camera mounted on the end-effector would improve on both counts.

Moved to look behind the target when the arm stalls, it could provide the re-observation without fixed side cameras, and a grasp re-selected from that view could close the remaining gap to always-on coverage (§6.2).

Grasp success itself saturates near 60% because objects are pushed out during the close, and none of the proposer, re-planning or close modifications we tried raised it. The natural next step is a deterministic, higher-fidelity physics engine such as GPU PhysX.

For limitations, all results were obtained in simulation. Contacts in CPU PhysX are not deterministic across scene restores, so effects of a few scenes remain within noise, and GG-CNN2 was run with pretrained weights on out-of-distribution input, so its success rate is a lower bound. Code, raw trials and a table-to-script map are in the README of the submitted repository. The video is submitted along with this report or it can be accessed online through `https://youtu.be/EwMxesAZIOY?si=MK7WUhrrUA-1QTWO`

8 AI Usage Declaration

Large language model (LLM), generative AI, was utilised in this work. Anthropic's Claude Sonnet 5.5 was used to implement the proposed controller scheme and its baselines. It was also used to schedule large-scale experimentation, assist writing report, and create nice visualisations.

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